INFINITE JETS

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1. Fibre bundles—2. The Peetre–Slovák Theorem—3. Sheaves—4. Differential operators—5. The proof.

1. Fibre bundles

Consider a smooth manifold M. (All manifolds will be smooth.) Suppose we have another manifold E with a map $\pi\colon E \to M$. If the preimages of π at each point $p \in M$ all look the same, then we say that E is a fibre bundle over M. Notice that π must be surjective, unless we're being silly.

Example 1.1. The cotangent bundle T^*M is a fibre bundle. So are $\Lambda^k T^*M$ for each k.

Given a manifold F, the product $M \times F$ is a fibre bundle over M; here F is the fibre.

The Möbius band is a fibre bundle over S^1 with fibre $\mathbb{R}$.

A section of a fibre bundle $E \xrightarrow{\pi} M$ is a smooth map $M \xrightarrow{s} E$ such that $M \xrightarrow{s} E \xrightarrow{\pi} M$ is the identity.

Example 1.2. A section of $\Lambda^k T^*M \to M$ is a differential k-form on M.

A section of $M \times F \to M$ is nothing but a map $M \to F$.

Given a fibre bundle E, write $\mathcal{E}(M)$ for the set of all sections of E.

Remark 1.3. $\mathcal{E}(M)$ is actually some kind of infinite-dimensional manifold. We won't address this point because it would take too long and I don't want the message of the talk to be buried in language.

2. The Peetre–Slovák Theorem

Here is the focus of the next forty minutes:

Theorem 2.1. *Let E, E' be fibre bundles on a manifold M. Let $D\colon \mathcal{E}(M) \to \mathcal{E}'(M)$ be a smooth map between their sets/manifolds/whatever of sections. The following are equivalent:*

(i) D is a differential operator.
(ii) D is local (it comes as the global sections of a morphism of sheaves $\mathcal{E} \to \mathcal{E}'$).

If you don't understand the statement, don't worry: this is the main point of the talk. If you do, then the statement should be surprising. Differential operators have to do with derivatives, and sheaves only know about locality.

Example 2.2 (Lagrangian field theory)**.** In the setup of Lagrangian field theory, we have a spacetime manifold M, and we have classical fields (which are sections of some fibre bundle $E \to M$) and a Lagrangian density, which is a (possibly nonlinear) differential operator

$$L\colon \mathcal{E}(M) \to \Omega^d(M).$$

(The fibre bundle E' in this case is $\Lambda^d T^*M$.) By the theorem, this is the same as a morphism of sheaves

$$\mathcal{L}\colon \mathcal{E} \to \Omega^d$$

where Ω^d is the sheaf of differential d-forms on M. Thus, Peetre–Slovák tells us that the dependence on the fields we ask of the Lagrangian encodes locality, the main principle of field theory.

3. Sheaves

Given a fibre bundle $E \to M$, we can restrict our attention to what happens at an open subset $U \subseteq M$. Write $\mathcal{E}(U)$ for the set of sections of $E|_U$. Here are two facts:

- Given $U \subseteq V$ two open subsets of M, I can restrict sections from V to U: this gives
$$\text{restrict}\colon \mathcal{E}(V) \to \mathcal{E}(U)$$
- If I have an open subset U and an open cover $U = \bigcup_i U_i$, then a point in $\mathcal{E}(U)$ is the same as a point in each $\mathcal{E}(U_i)$ which agree when restricted to $\mathcal{E}(U_i \cap U_j)$.

This is what it means to say that $\mathcal{E}$ is a sheaf; it's the "sheaf of sections" of the fibre bundle E.

Definition 3.1. A "morphism of sheaves" $\phi\colon \mathcal{E} \to \mathcal{E}'$ is a smooth map $\phi(U)\colon \mathcal{E}(U) \to \mathcal{E}'(U)$ for each $U \subseteq M$, which respects restriction to open subsets.

Example 3.2. A bundle map $f\colon E \to E'$ determines a morphism of sheaves $\mathcal{E} \to \mathcal{E}'$ which sends a local section $s\colon U \to E$ to $f \circ s\colon U \to E'$.

Not all morphisms of sheaves are of this form: we will see the main non-example in the next section.

4. Differential operators

Heuristically, a map $D\colon \mathcal{E}(M) \to \mathcal{E}'(M)$ is a *differential operator* if $D(s)$ depends smoothly on the value of a section s and its partial derivatives. To formalise this heuristic, we need some language:

Definition 4.1 (Germs)**.** A "germ of a section at $x \in M$" is, informally, how a section looks in an arbitrarily small neighbourhood of x. The set of germs of sections at x is:
$$\mathcal{E}|_x \coloneqq \frac{\{(U,s) : U \text{ an open neighbourhood of } p,\ s \in \mathcal{E}(U)\}}{(U_1,s_1) \sim (U_2,s_2) \iff \exists (U_3,s_3) \text{ s.t. } U_3 \subseteq U_1 \cap U2 \text{ and } s_1|_{U_3} = s_2|_{U_3} = s_3}$$
is the "infinite-dimensional manifold" of germs of sections at x. (The stalk at x.)

Definition 4.2. Given $k \in \mathbb{N}$, there is a fibre bundle $J^k E \to M$. The fibre of $J^k E$ at $x \in M$ is
$$\frac{\mathcal{E}|_x}{[U_1,s_1] \sim [U_2,s_2] \iff s_1, s_2 \text{ have the same partial derivatives of at } x \text{ up to total order } k}$$
is the set of germs of sections of E, modulo the equivalence relation that all their partial derivatives up to order k are equal. It naturally has the structure of a finite-dimensional manifold.

Example 4.3. $J^0 E = E$. $J^1 E$ is the set of all possible tangent planes of sections of E.

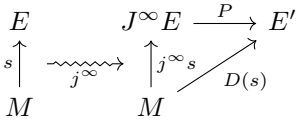
Remark 4.4. A section of $J^1 E \to E$ is the same thing as a connection on E.

If two germs have the same partial derivatives up to order $(k+1)$, then in particular they have the same partial derivatives up to order k. There are bundle maps $J^{k+1}E \to J^k E$ which simply forget the $(k+1)$-order data. These forgetful maps define a sequence
$$\cdots \to J^{k+1}E \to J^k E \to \cdots \to J^2 E \to J^1 E \to E,$$
and $J^\infty E$ is the limit of this sequence, in some appropriate sense. (Fréchet manifold, locally pro-manifold, smooth sets…) A point of $J^\infty E$ is like an abstract Taylor series of a section $M \to E$ at a point of M.

Definition 4.5 (Jet prolongation)**.** Write $\mathcal{J}^\infty \mathcal{E}$ for the sheaf of sections of $J^\infty E$. There is a morphism of sheaves $j^\infty\colon \mathcal{E} \to \mathcal{J}^\infty \mathcal{E}$ which sends a local section $s \in \mathcal{E}(U)$ to its *jet prolongation* $j^\infty s \in \mathcal{J}^\infty \mathcal{E}(U)$. (I won't construct it; ask me at lunch if you want.) The jet prolongation $j^\infty s$ records not just the value of a section s at each point of U, but also all its higher partial derivatives at each point of M.

Remark 4.6. The morphism $j^\infty\colon \mathcal{E} \to \mathcal{J}^\infty\mathcal{E}$ definitely *does not* come from a bundle map $E \to J^\infty E$. The only such bundle map I can think of is the one which sends a point $e \in E$ to the jet $(e, 0, 0, \dots)$. The associated sheaf morphism is of a completely complementary nature from the jet prolongation morphism.

Definition 4.7. A smooth map $D\colon \mathcal{E}(M) \to \mathcal{E}'(M)$ is a *differential operator* if it is of the form $D(s) = P(j^\infty s)$ for some bundle map $P\colon J^\infty E \to E'$.



If such a P exists, it's unique. Sometimes people call P itself the differential operator. The differential operator is of degree k if P factors through the projection $J^\infty E \to J^k E$.

Corollary 4.8 (Structure theorem for sheaf morphisms)**.** *It is a surprising consequence of the Peetre–Slovák theorem that the jet prolongation is the universal non-example of a morphism induced by a bundle map: every morphism $\mathcal{E} \to \mathcal{E}'$ factors as j^∞ followed by a bundle map.* □

5. The proof

The map from differential operators to morphisms of sheaves is

$$(P\colon J^\infty E \to E') \mapsto (\phi_P\colon \mathcal{E} \to \mathcal{E}')$$

where ϕ_P sends a local section $s\colon U \to E$ to $U \xrightarrow{j^\infty s} J^\infty E \xrightarrow{P} E'$.

The map from the sheaf side to the differential operator side is more difficult.

$$(\phi\colon \mathcal{E} \to \mathcal{E}') \mapsto (P_\phi\colon J^\infty E \to E').$$

To define P_ϕ, we will represent a point of $J^\infty E$ as $j^\infty_x s$ where s is some local section of E.

Lemma 5.1. *The map P_ϕ is given by*

$$P_\phi(j^\infty_x s) = \phi(s)(x).$$

This is independent of the choice of s used to define the point $j^\infty_x s \in J^\infty E$.

Proof. The thing you need to show is: If $s, s' \in \mathcal{E}(U)$ are sections defined on a neighbourhood U of $p \in M$, whose ∞-jets agree at $x \in M$, then the sections $\phi(s), \phi(s') \in \mathcal{E}'(U)$ have the same value at x. This is strangely difficult: it relies on a deep theorem called Whitney's Extension Theorem. □

Then, one shows that $P \mapsto \phi_P$ and $\phi \mapsto P_\phi$ are mutual inverses. The other main ingredient is:

Lemma 5.2 (Local finiteness of order)**.** *Any point $\underline{x} \in J^\infty E$ has an open neighbourhood V such that $P_\phi|_V$ factors through $J^\infty E \to J^n E$.*

Proof. The proof also invokes Whitney's Extension Theorem. □

All of the details are in the preprint *Peetre–Slovák's theorem revisited* by Navarro and Sancho.